# Spectral and Temporal Studies of the Gamma-ray Blazar B2 1811+31


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## 1. Background

Galaxies emitting from their cores, powered by growing **supermassive black holes** (SMBHs), can have bright central regions called **Active Galactic Nuclei** (AGN). When AGN have jets of highly energetic plasma pointed nearly directly at Earth, they are called blazars**.** In the core, accretion of matter powers the relativistic outflows in the jets. One reason blazars are intriguing is they show rapid **variability** in flux.

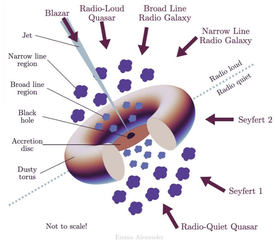



Figure 1. Artist's Scheme for AGN unification

**Spectral energy distributions** (SEDs) reveal information on the configuration and processes driving sources' emission.

- ★ SEDs help **develop models** to describe variability.
- ★ **Two** broad, non-thermal radiation **components** in blazars' SEDs.
- ★ **Lower-energy** (Infrared to UV) emission is **well understood** and attributed to synchrotron emission.
- ★ The source of the **higher-energy** (X-ray to gamma-ray) component is still **unclear**.

## 2. Motivation

In October 2020, Fermi-LAT recorded **high energy** (HE) detections from the location of B2 1811+31, a source with an otherwise significantly lower flux. The source was determined to be in an elevated state, and, after an alert from MAGIC Telescope, VERITAS pointed its telescopes to the source, receiving significant detections.

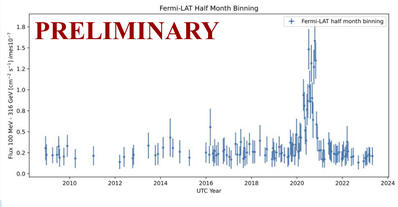



Figure 3. Fermi Lightcurve of B2 1811+31

## 3. Goals

- ★ Create a gamma-ray SED for the flare and quiescent states.
- ★ Fit to a spectral model that best describes the shape of the SED.

***Why?***

- ★ Investigating blazars' spectral shapes will help uncover the drivers behind blazar variability.
- ★ Are the different flux states in blazars a result of a singular phenomenon, or is there evidence of different emission processes?

## 4. Instruments

### VERITAS Gamma-Ray Observatory



Figure 2a. VERITAS (credit: VERITAS)

- ★ Ground-based array of 4 gamma-ray telescopes
- ★ Uses Cherenkov light in atmospheric interactions to record gamma-ray events.
- ★ Effective energy range ~ 85 GeV - 30 TeV

### Fermi Large Area Telescope (Fermi-LAT)

- ★ Gamma-ray survey satellite
- ★ ~ 3 hrs to survey the sky
- ★ Effective range ~ 20 MeV to >300 GeV

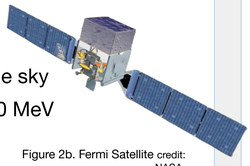


Figure 2b. Fermi Satellite credit: NASA

## 5. Software

### VEGAS

- ★ Analysis package designed to analyze telescope data from VERITAS.
- ★ Soft cuts: cover a wide range of energies with the lowest energy threshold.

### Fermipy

- ★ Software for analysis of gamma-ray data collected by the Fermi-LAT telescope.

### Gammapy

- ★ Allows joint fits of multi-instrument data.
- ★ Useful for comparing analyses like the flaring and quiescent –or latent– states.

## 6. Results & Discussion

B2 1811+31 is classified as an **Intermediate Peaked BL Lac object (IBL)**, peaking between high to low energies, at z = 0.117. For constant emitting sources, **significance is expected to grow linearly** as the square root of time. This growth factor (gf) serves serves to compare the elevated and quiescent states.

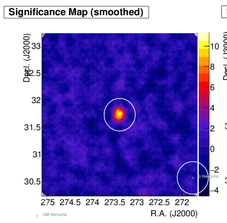



Figure 4a. Flare Significance Map

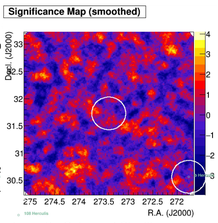



Figure 4b. Quiescent Significance Map

**VERITAS observations**:

- ★ Flare: 10/15/2020 - 10/19/2020
  - gf = 4.11 (9.18 σ in 4.33 hrs)
- ★ Quiescent: 04/19/2023 - 06/27/2023
  - gf = 0.09 (0.18 σ in 3.38 hrs)

***Detection Timeline***

Fermi-LAT light curves were used to statistically define the periods of interest in 1-day and 0.5-month bins.

- ★ Flare: 10/17/2020 - 11/04/2020
- ★ Quiescent: 01/16/2021 - 07/05/2023

Flare definition is poorly defined, working to reach a better understanding of the relevant periods of interest for this source.

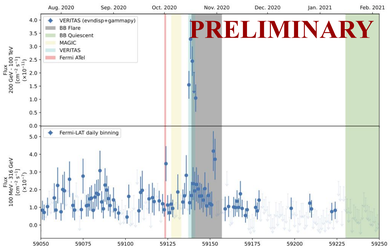



Figure 5. Time Annotated Lightcurve

***Model Fitting***

- ★ **Power law (PWL)**: slow decline in flux.
- ★ **Exponential cutoff power law (EXPWL)**: exponential decay at higher energy cutoff.
- ★ **Log parabola (LP)**: curve and soft drop.

| Period | PWL | EXPWL | LP |
|---|---|---|---|
| Flare | 32.68 | 12.26 | 8.21 |
| Quiescent | 5.03 | 4.32 | 4.4 |

Table 1. BIC measures wellness of fit of Models

Extragalactic Background Light (EBL) scatters and absorbs light, dimming the observed flux, so we account for absorption in all fits. To compare the fitting among models with different parameters, we use Bayesian Information Criterion (BIC). The closer the BIC are to 0, the better the fit.

- ★ Currently, the **flare** is best fit to an **LP** and the **quiescent** to an **EXPWL**.

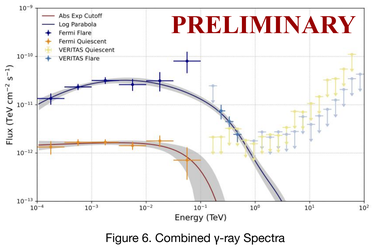



Figure 6. Combined γ-ray Spectra

## 7. Takeaways & Next Steps

- ★ Complete rise and decay time analysis of the peak, or peaks, to increase precision of flare definition, and consider Markov Chain Monte Carlo (MCMC) algorithm to fit the SED.
- ★ Combine low-energy data for broadband SED!

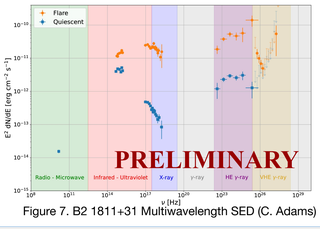



Figure 7. B2 1811+31 Multiwavelength SED (C. Adams)

## 8. References


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